

Building a Reproducible Scholarly Search Toolkit

Episode 16b: Verification & Hallucination Detection

Nexus Scholar Suite

The LLM Hallucination Threat

Synthetic Citations

LLM agents frequently invent fake DOIs, blend distinct paper titles, and generate phantom authors.



DocumentVerifier API

```
verifier = DocumentVerifier()

# 1. Verify Document Integrity
result = verifier.verify_document(doc)
if result.is_verified:
    print(f"Verified via DOI: {result.matched_doi} (Conf: {result.confidence:.2f})")
else:
    print(f"Flagged Hallucination: {result.issues}")

# 2. Hydrate Missing Metadata
enriched_doc = verifier.hydrate_metadata(doc)
```